

REVIEW

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Mapping waste mediated zoonotic and antimicrobial resistance risks in rural One Health systems

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Abstract

Rural waste systems represent important interfaces linking human, animal, and environmental health. Poor management of livestock manure, wastewater, and other solid and liquid waste streams has been associated with increased risks of zoonotic disease transmission and antimicrobial resistance (AMR). However, evidence from rural and low-resource settings remains fragmented. This study synthesized existing evidence on zoonotic and AMR risks within the waste–livestock–human–environment nexus and developed a conceptual framework to support integrated risk management in rural One Health systems. A narrative synthesis was conducted using 93 peer-reviewed studies published between 2000 and 2025. Studies were identified through major scientific databases and screened using predefined inclusion criteria. Data were extracted and organized thematically according to waste type, exposure pathways, geographic context, biological hazards, and reported interventions. The findings indicate that animal manure, human faeces, wastewater, and slaughter-facility effluent were the waste streams most frequently associated with zoonotic and AMR risks. Bacterial zoonoses and resistant pathogens were reported in approximately 72% of the studies. Fewer than 10% of studies applied quantitative microbial risk assessment or molecular AMR surveillance approaches, highlighting major analytical and monitoring gaps. Frequently reported risk factors included poor hygiene practices, unsafe water use, limited infrastructure, and weak regulatory enforcement, particularly in rural African and South Asian settings. Based on the synthesis, a Rural Waste-Mediated One Health Risk Prioritization Framework was developed to support coordinated surveillance, policy development, and community-level interventions in rural systems.

Keywords Environmental contamination, Livestock health, One Health, Rural waste management, Zoonotic diseases



1 Introduction

Solid and liquid waste management remains a major environmental and public health challenge in many rural areas of developing countries. Globally, approximately 2.24 billion people, most of whom live in rural and peri-urban regions of low- and middle-income countries, lack access to basic waste collection services [30]. Unlike urban areas that increasingly benefit from improved waste infrastructure, many rural communities continue to depend on informal waste disposal practices such as open dumping, open burning, and the discharge of untreated liquid waste into surface water bodies [61, 84]. These practices contribute to environmental pollution and negatively affect livestock health, food safety, water quality, and ecosystem functioning [1, 8, 72].

Rural waste-mediated One Health systems involve interconnected interactions among humans, animals, waste streams, and the environment within shared ecosystems. In many rural agricultural communities, livestock are exposed to contaminated water sources, untreated wastewater, poorly managed household waste, and accumulated animal manure, creating favorable conditions for zoonotic disease transmission and antimicrobial resistance (AMR) emergence [32, 66, 88]. In this review, rural settings refer to agricultural and resource-constrained communities characterized by livestock-dependent livelihoods, limited waste management infrastructure, and close human–animal–environment interactions [30, 47, 87]. These conditions increase the risk of food contamination, environmental degradation, zoonotic disease transmission, and dissemination of antimicrobial resistance determinants across One Health domains [21, 32, 68]. Rural waste-mediated systems therefore represent an important but underexplored pathway through which environmental contamination influences livestock health, food safety, zoonotic hazards, and antimicrobial resistance risks in rural settings [26, 66].

Rural communities often face major structural and institutional challenges that limit effective waste management. Waste collection systems, drainage infrastructure, and regulatory enforcement mechanisms are frequently inadequate or absent [10, 46, 47]. At the same time, changing consumption patterns, agricultural intensification, and increased use of packaged goods have introduced new waste streams such as plastics, batteries, chemical containers, and electronic waste into rural environments [22]. Although rural areas generate a smaller proportion of total solid waste (20–30%), they may experience disproportionate impacts due to limited awareness, constrained institutional capacity, and a high dependence on livestock and agriculture for livelihoods [87]. Consequently, rural waste streams are becoming increasingly complex and hazardous, increasing the potential for environmental contamination and associated health risk [18, 36].

In livestock-based rural economies, unmanaged waste streams create multiple exposure pathways for animals and humans. Livestock such as cattle, goats, and poultry often scavenge near households and open dumping sites, where they may ingest plastics, contaminated food materials, or toxic substances [6, 66]. Studies conducted in Ethiopia and India have reported that up to 35% of cattle and goats ingest non-biodegradable materials, including plastic bags and pesticide packaging, resulting in gastrointestinal blockages, reduced productivity, and mortality [43, 67]. Toxic contaminants such as heavy metals, dioxins, and antibiotic residues have also been detected in animal products, raising concerns related to food safety and chronic human exposure [3, 68, 72]. During rainfall events, runoff from unregulated waste disposal sites and open defecation areas can transport pathogens, heavy metals, and agrochemical residues into grazing lands and

water sources, further increasing exposure risks for both animals and humans [15, 57]. Despite these interactions, awareness of links between environmental contamination and livestock health remains limited among smallholder farmers in many settings [26].

Improper waste management in rural areas is also increasingly recognized as relevant to the global challenge of antimicrobial resistance (AMR). The disposal of animal effluent, human sewage, and pharmaceutical residues into the environment can create reservoirs in which antimicrobial resistance genes are maintained and exchanged among microbial populations [32, 81, 88]. In rural livestock systems, antibiotics used in veterinary and human medicine may persist in soil and water, facilitating selection and dissemination of resistant microorganisms that can re-enter the food chain. These processes reinforce the importance of the One Health framework, which recognizes the interconnectedness of human, animal, and environmental health [21]. However, despite growing recognition of these interactions, research and policy responses remain fragmented, and integrated conceptual approaches for risk prioritization remain limited.

This review synthesizes current evidence on waste-related zoonotic and antimicrobial resistance risks in rural livestock systems and proposes a Rural Waste-Mediated One Health Risk Prioritization Framework. The framework aims to support future research, surveillance, and policy development by identifying major waste streams, exposure pathways, biological hazards, and intervention gaps. The review further highlights the importance of integrated and context-specific waste management approaches for strengthening environmental protection, livestock health, food safety, and public health in rural communities. The review also addresses Sustainable Development Goals (SDGs 6, 12, and 13).

2 Methodology

2.1 Study design

This study employed an integrative narrative review approach to synthesize evidence on zoonotic disease risks and antimicrobial resistance (AMR) pathways occurring at the rural waste–livestock–human–environment interface. The review aimed to examine how waste-related exposure pathways influence livestock health, environmental contamination, zoonotic disease transmission, and AMR emergence within rural One Health systems. An integrative narrative review design was selected because the available evidence spans multiple disciplines, including epidemiology, environmental health, livestock production, microbial contamination, antimicrobial resistance surveillance, and policy analysis. The approach allowed the synthesis of evidence from different study designs while maintaining structured procedures for study identification, screening, and evidence interpretation.

2.2 Data sources and search strategy

The literature search included peer-reviewed studies and eligible grey literature published between January 2000 and March 2025. This period was selected to capture growing global attention toward One Health systems and environmental antimicrobial resistance. Electronic searches were conducted using PubMed/MEDLINE, Scopus, Web of Science, and Google Scholar. Additional studies were identified through manual screening of reference lists from eligible publications. Search terms were organized around four major domains which included waste-related factors, livestock production

systems, health outcomes, and One Health contextual determinants. Waste-related search terms included rural waste management, livestock manure, wastewater, slaughter waste, environmental contamination, solid waste, household waste, greywater, and waste exposure pathways. Livestock-related terms included livestock, cattle, goats, poultry, dairy systems, and animal production. Health outcome terms included zoonotic disease, zoonoses, antimicrobial resistance, AMR, food safety, pathogen transmission, and antimicrobial resistance genes. System-level terms included One Health, rural systems, environmental health, agricultural communities, and low-resource settings.

2.3 Eligibility criteria

Studies were considered eligible for inclusion if they were published between 2000 and 2025 and examined rural or peri-urban agricultural settings involving livestock-associated waste exposure pathways. Eligible studies investigated solid waste, livestock manure, wastewater, slaughter waste, greywater, environmental contamination, or related waste streams with implications for livestock health, zoonotic disease transmission, antimicrobial resistance emergence, food safety risks, or environmental contamination. Studies addressing livestock–human–environment interactions relevant to One Health systems were prioritized. Priority was given to epidemiological studies, environmental surveillance investigations, microbial contamination studies, AMR surveillance studies, intervention evaluations, systematic reviews, and policy analyses relevant to rural One Health systems. Only English-language publications and eligible grey literature were included.

2.4 Exclusion criteria

Studies were excluded if they focused exclusively on urban systems without livestock relevance, lacked waste-associated exposure pathways, reported insufficient methodological information, or were unavailable in full text. Duplicate publications identified during screening were removed. Conference abstracts without complete reports were excluded. Laboratory-based microbiological investigations lacking environmental or livestock interface relevance were not considered for synthesis. Opinion articles and editorials lacking empirical evidence or conceptual contribution to rural waste–livestock–human–environment interactions were also excluded.

2.5 Study selection process

Study identification and screening procedures were informed by Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) reporting principles to strengthen methodological transparency. Database searching initially identified 426 potentially relevant records. Following duplicate removal, 359 unique publications underwent title and abstract screening against predefined eligibility criteria. Publications that did not address waste-associated livestock exposure pathways, zoonotic risks, antimicrobial resistance determinants, or One Health interactions relevant to rural agricultural systems were excluded during preliminary screening. Full-text eligibility assessment was conducted for 138 studies considered potentially relevant following abstract screening. During detailed review, studies were excluded where methodological information was insufficient, livestock relevance was absent, exposure pathways were not clearly characterized, or findings fell outside the review scope. Following eligibility

assessment, 93 studies satisfied inclusion criteria and were retained for integrative synthesis. The study identification and selection process is summarized in the PRISMA flow diagram presented in Fig. 1.

2.6 Data extraction and synthesis

Data extraction was done using a structured framework developed to capture study characteristics and major evidence domains relevant to rural waste-associated health risks. Extracted variables included geographical location, livestock production characteristics, waste stream categories, livestock exposure pathways, zoonotic hazards identified, antimicrobial resistance determinants, environmental contamination routes, intervention approaches, governance dimensions, and reported health outcomes. Evidence synthesis followed thematic analytical procedures to identify recurring exposure mechanisms, contamination pathways, intervention opportunities, and evidence gaps across studies. Particular emphasis was placed on understanding interactions linking waste generation, environmental contamination, livestock exposure, zoonotic disease emergence, food safety risks, and antimicrobial resistance development within rural One Health systems. Risk levels were derived through qualitative evidence synthesis based on the frequency, consistency, and reported severity of zoonotic and antimicrobial resistance (AMR) exposure pathways across the reviewed literature.

2.7 Review limitations

This review has several limitations. Restricting inclusion to English-language publications may have excluded relevant evidence from other settings. Methodological differences across studies limited direct comparison between findings. Variability in waste characterization, livestock exposure assessment, and antimicrobial resistance surveillance approaches also introduced heterogeneity across the evidence base. Structured screening procedures, predefined eligibility criteria, and quality appraisal methods were applied to strengthen transparency and reduce potential bias.

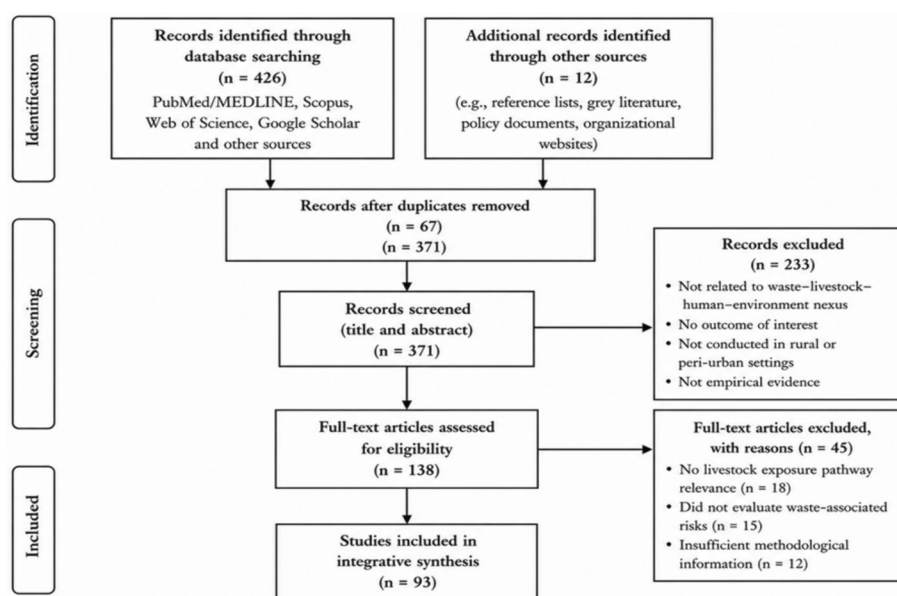


Fig. 1 PRISMA flow diagram illustrating literature identification, screening, eligibility assessment, and study inclusion procedures

3 Results and synthesis

3.1 Characteristics of included studies

The ninety-three studies included in this review covered a wide range of geographic contexts across Asia, sub-Saharan Africa, Europe, and the Americas. The dataset comprised peer-reviewed research articles as well as eligible grey literature sources that provided sufficient detail for thematic characterization of waste streams, exposure pathways, and associated health outcomes. Four studies adopted a global perspective, while the majority were region- or country-specific. Studies from Zimbabwe, South Africa, Nigeria, Ghana, Ethiopia, Uganda, and Tanzania were among the most represented study settings, while a smaller number of studies originated from the United States and other global contexts (Fig. 2).

Table 1 presents the geographic scope and classification of the 93 studies included in the review. Most studies were global or multi-country in scope, followed by country-specific studies conducted primarily in Africa and Asia. Smaller proportions included regional, continental, and comparative studies.

Most studies focused on animal manure, wastewater, or effluent from slaughter facilities as the primary waste streams of interest. A smaller number of studies also examined biosolids, greywater, and waste milk as additional sources of environmental and livestock exposure. Zoonotic pathogens were investigated in several studies, with frequent attention given to *Salmonella*, *Campylobacter*, *Escherichia coli* (*E.coli*), *Cryptosporidium*, *Giardia*, *Leptospira*, and *Brucella* spp. Antimicrobial resistance was addressed in seven studies, which reported the detection of multidrug-resistant bacteria, including *E. coli*, *Staphylococcus*, *Enterococcus*, and *Salmonella*. Resistance to commonly used antimicrobials such as tetracyclines, sulfonamides, and vancomycin was documented, often in association with identifiable resistance genes. Environmental dissemination emerged as a recurring theme, particularly in studies examining wastewater effluent and biosolids as potential reservoirs and transmission pathways for antimicrobial resistance genes. Only a limited number of studies evaluated intervention measures, most of which focused on water, sanitation and hygiene (WASH) and farm-level biosecurity practices, with outcomes varying according to local socioeconomic and infrastructural conditions.

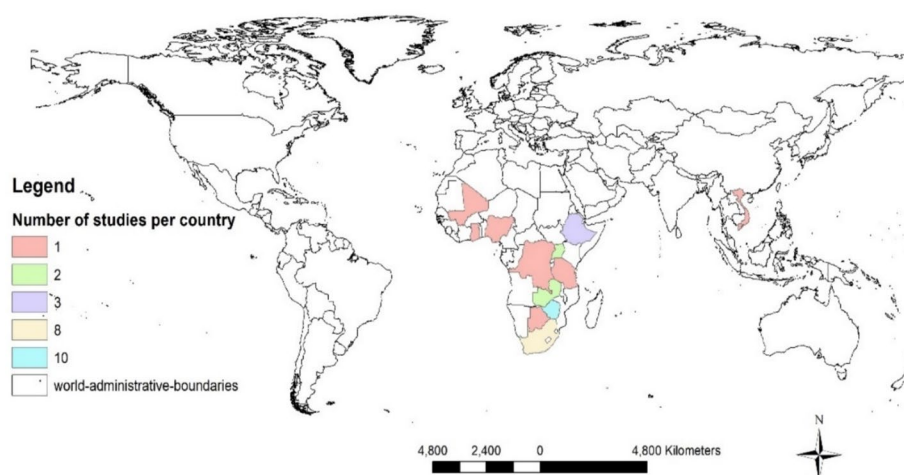


Fig. 2 Geographical distribution of included studies (n = 93)

Table 1 Geographic distribution and classification of studies included in the review (n = 93)

Category group	Description	Number of studies	Percentage (%)
Country-specific (Africa & Asia)	Studies focused on individual countries in Africa and Asia	33	35.5
Regional studies	Studies covering specific regions within Africa	9	9.7
Continental focus	Studies with an Africa-wide continental focus	5	5.4
Comparative (Europe & Africa)	Comparative studies between Europe and Africa	1	1.1
Comparative (Asia – Vietnam)	Comparative studies involving Asian contexts including Vietnam	1	1.1
Global/Multi-country/Non-country specific	Studies with global, multi-country, or non-country-specific scope	39	41.9
Total		93	100.0

3.2 The changing face of rural waste

Analysis of the included studies revealed consistent evidence of a transition in the composition and characteristics of rural waste streams. Across multiple geographic regions and study designs, the reviewed literature documented a shift from predominantly organic and biodegradable waste toward increasingly complex mixtures containing plastics, chemical containers, batteries, and electronic waste [22, 27, 47]. These patterns were reported across rural low- and middle-income settings, reflecting changes in consumption patterns, agricultural intensification, and market penetration of packaged goods [84, 87].

Several studies further highlighted that this transformation has rendered traditional disposal practices and policy frameworks increasingly ineffective, contributing to elevated environmental contamination and emerging toxicological risks [7, 36]. The volume of waste in rural African communities continues to pose significant challenges, with per capita generation estimated between 0.15 and 0.5 kg per person per day [2, 75]. More revealing, however, is the evolving composition of this waste. As summarized in Table 2, organic materials remain the dominant fraction (50–70%) but are increasingly accompanied by plastics (10–20%), which are rapidly proliferating. Of particular concern are hazardous components such as batteries, pesticide containers, and electronic waste, which, although constituting a minor proportion by weight (<1–3%), present disproportionate risks due to their persistence and toxicity [36]. Consequently, conventional disposal methods, including open burning and shallow burial, are no longer merely inadequate; they actively introduce dioxins, heavy metals, and other toxic substances into the environment [7].

Composition varies widely based on local socio-economic factors, consumption patterns, and seasonal agricultural cycles. Organic waste dominates rural waste streams due to subsistence farming and low processed food consumption. Plastics are increasing rapidly due to market penetration, even in remote rural areas. Hazardous waste is often underestimated but poses emerging risks with increasing rural electrification and the use of agrochemicals.

Table 2 Categories and typical composition of solid waste in rural areas of developing countries

Waste category	Description	Approximate % composition (by weight)	Examples	References
Organic Waste	Food scraps, crop residues, animal manure	50% – 70%	Vegetable peelings, leftover food, livestock manure	Mawede [41], Okuthe [64]
Plastics	Packaging materials, bags, sachets, containers	10% – 20%	Plastic bags, sachets, bottles	Farai & Jerie [18], Mazhandu et al. [42]
Paper & Cardboard	Newspapers, packaging, cartons	5% – 12%	Newspapers and packaging waste	Mofokeng et al. [48], Mofokeng et al. [49]
Metals	Cans, scrap metal, tins	1% – 5%	Beverage cans, tins	Gomaa et al. [24], Mamman [38]
Glass	Bottles, broken glass	1% – 4%	Glass bottles from beverages	Ezeudu & Kennedy [17]
Hazardous Waste	Batteries, e-waste, pesticides	< 1% – 3%	Discarded batteries, solar panel parts	Lebbie et al. [36]
Other/Inert	Soil, ash, construction debris	5% – 10%	Ash from burning, dirt, rubble	Krishna [35], Ramli et al. [71]

A similar trajectory of increasing complexity is observed in rural liquid waste streams (Table 3). Previously dominated by rainwater and basic greywater, these streams now include untreated household greywater, blackwater from pit latrines, and agricultural runoff enriched with pesticides and fertilizers [33, 53, 70, 73]. This convergence creates a substantial contamination risk for both surface and groundwater, which often serve as primary water sources for livestock, establishing direct pathways for exposure to pathogens and chemical hazards [32, 93].

3.3 Pathways of exposure

The escalation of rural waste from predominantly benign to hazardous would pose a limited problem if it remained confined. However, within the tightly interwoven ecosystems of rural communities, such contamination inevitably enters the food chain. Exposure occurs through multiple, intersecting pathways, with livestock, often free-ranging and scavenging, serving both as primary recipients and vectors of environmental pollutants. Across numerous villages in developing regions, animals navigate landscapes increasingly fraught with hidden hazards, transforming routine feeding into a significant health risk. Figure 3 provides a conceptual model of this contamination, illustrating the complex network linking waste sources to animals and, ultimately, human hosts. The evidence indicates that exposure is not episodic but continuous, with several principal routes emerging.

The most immediate pathway is through grazing near waste disposal sites and human settlements. In the absence of formal collection systems, households frequently discard refuse in open spaces that double as grazing grounds. For example, in Zimbabwe's Lupane and Gwanda districts, over 70% of households dispose of solid waste openly, and 42% of goat owners report their animals ingesting plastics or food scraps from these sites [12, 59]. Such ingestion can result in fatal intestinal blockages, while sharp objects pose risks of internal injury [4, 5].

Another critical pathway involves the consumption of contaminated forage and water. Crops irrigated with untreated greywater or cultivated near sewage outlets may harbor

Table 3 Categories and characteristics of liquid waste in rural areas of developing countries

Liquid waste category	Description	Common sources	Environmental and health concerns	References
Greywater	Wastewater from domestic activities (bath, laundry, kitchen)	-Household activities	-Contains organic matter, detergents, nutrients; can contaminate soil and surface water if untreated	WHO & UNICEF [90]; Okafor et al. [62]
Blackwater	Wastewater containing human excreta from toilets, pit latrines	-Sanitation facilities, open defecation	-Pathogens (bacteria, viruses, parasites); contamination of water sources causing diarrheal diseases	Sobsey et al. [78], Hosain et al. [29]
Animal Effluent	Wastewater from livestock (urine, feces, wash water)	-Livestock pens, kraals, grazing runoff	-High organic load, pathogens, nutrients contributing to eutrophication	Mergia et al. [43], Zahoor & Mushtaq [93]
Agricultural Runoff	Water carries fertilizers, pesticides, and sediments from fields	-Crop fields, small-scale farming	-Pesticide residues and nutrient loading, causing water pollution and toxicity	Mutika et al. [53], Gomaa et al. [24]
Industrial and Mining Wastewater	Liquid wastes from small-scale industries, mining activities	-Informal mining, local processing units	-Heavy metals, chemicals contaminating surface and groundwater	Narayan et al. [54], Mihai et al. [47]
Stormwater Runoff	Rainwater runoff carrying various pollutants	-Urban and peri-urban agricultural areas	-Carries litter, pathogens, and chemicals into water bodies	WHO & UNICEF [90]; Okafor et al. [62]

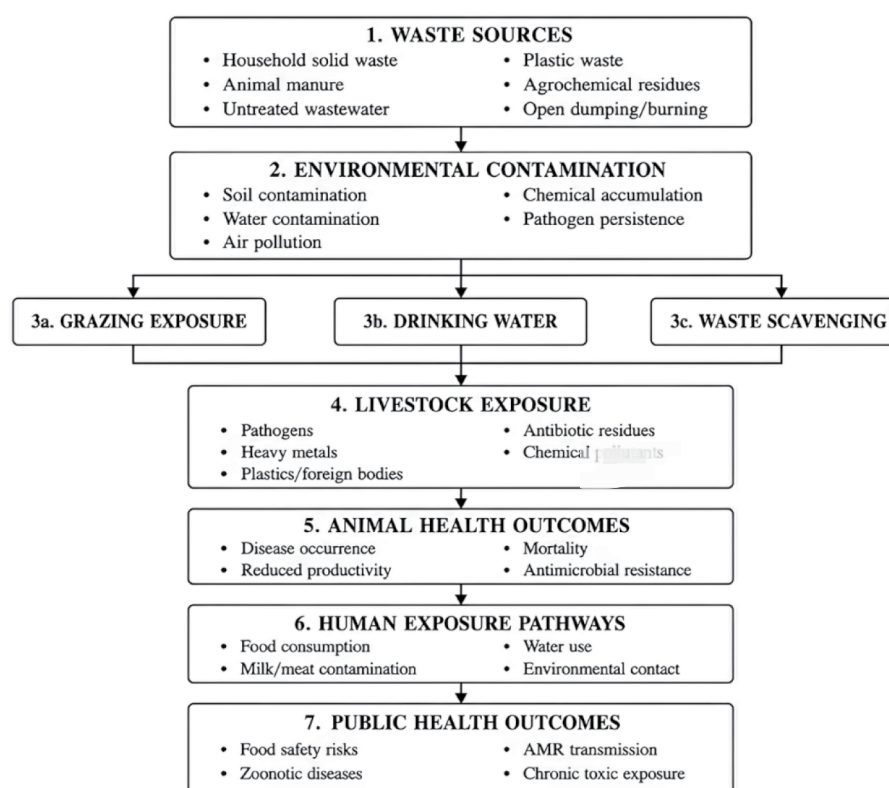


Fig. 3 Pathways to livestock contamination

pathogens and heavy metals [80]. During dry periods, livestock often rely on stagnant greywater or polluted surface water, significantly increasing exposure to diarrheal pathogens and infections such as leptospirosis [76]. This liquid-mediated route demonstrates that even areas lacking visible solid waste remain vulnerable to contamination. Exposure also occurs through direct contact and inhalation. Livestock traversing kraals where effluent accumulates are prone to dermatitis and hoof infections [39, 91]. Additionally, the open burning of plastics and rubber, a common disposal method, releases dioxins and fine particulate matter, which animals may inhale, resulting in chronic respiratory distress [7, 25]. Table 4 systematically summarizes these exposure routes, ranking both their likelihood and potential impact. Ingestion of waste and contaminated forage or water emerges as the most frequent and consequential pathway.

These direct exposures are further exacerbated by vector-mediated transmission, whereby flies and rodents attracted to waste act as carriers for pathogens such as *E. coli* and *Salmonella*, facilitating their spread to animals and animal products [55, 66]. Nonetheless, several communities are adopting adaptive practices that interrupt these exposure pathways. Measures such as fencing off waste pits, restricting livestock access to contaminated water sources, and pre-treating organic refuse before use as feed demonstrate that exposure is not unavoidable [28]. These interventions highlight actionable points for mitigating contamination and form the foundation for strategies discussed in subsequent sections addressing the consequences and solutions of rural waste-livestock-human interactions.

Table 4 Major pathways of livestock exposure to waste in rural communities of developing countries

Exposure pathway	Waste source	Commonly affected livestock	Reported effects	Likelihood of occurrence	Citation
Grazing near open dumps	Mixed household waste	Goats, cattle	Ingestion of plastics, digestive blockages, and internal injuries	Very High	Ntwali et al. [59], Priyanka & Dey [67]
Contaminated forage/water	Greywater, sewage, pesticide runoff	Goats, cattle	Diarrhea, parasite infections, and heavy metal accumulation	High	Zahoor & Mushtaq [93], Jimenez et al. [32]
Contact with contaminated soil	Greywater pools, kraal effluent	Cattle, pigs	Hoof rot, dermatitis, skin infections	High	Radingoana [69], Jimenez et al. [32]
Inhalation of smoke	Burning plastics, rubber	All livestock	Respiratory irritation, chronic stress	Moderate	Chavan et al. [7], Gordon [25]
Feeding on refuse	Market waste, unsorted kitchen waste	Goats, pigs	Food poisoning, ingestion of non-digestible	High	Riaz et al. [72], Ntwali et al. [59]
Vector-mediated transmission	Organic waste, faeces	Poultry, goats	Transmission of <i>E. coli</i> , <i>Salmonella</i> , and cryptosporidiosis	Moderate to High	Penakalapati et al. [66], WHO & UNICEF [90], Ncube [55]

Table 5 Health impacts of waste exposure on livestock

Exposure type	Health effect	Productivity impact	Likelihood of occurrence	Citations
Ingestion of plastics	Gastrointestinal obstruction, impaction	Weight loss, mortality	High	Ntwali et al. [59], Priyanka & Dey [67]
Contaminated water	Liver/kidney damage from heavy metals	Reduced milk yield, organ failure	Moderate to High	Zahoor & Mushtaq [93], Jimenez et al. [32]
Skin/hoof contact	Dermatitis, foot rot, ulcerative lesions	Lameness, reduced mobility	Moderate	Radingoana [69], Martineau [39], Penakalapati et al. [66]
Bacterial pathogens	Salmonellosis, leptospirosis, mastitis	Abortions, mastitis, death	High	Penakalapati et al. [66], Musara & Kapungu [51]
Inhalation (burning)	Respiratory infections, coughing	Lower stamina, chronic weakness	Moderate	Chavan et al. [7], Gordon [25]
Toxin exposure (feed)	Hormonal disruption, poor fertility	Low reproduction, miscarriages	Low to Moderate	Barik et al. [6], Franco et al. [19]

3.4 From animal illness to human plate

The pathways of exposure described earlier translate environmental contamination into tangible impacts on both animal and human health. This section shifts focus from the mechanisms of waste transfer to the consequences for livestock and human food safety, highlighting a dual burden: immediate effects on animal health and livelihoods, and the silent contamination of the human food chain.

3.4.1 Burden on animal health and livelihoods

Livestock exposed to contaminated waste experience both acute and chronic health effects (Table 5). Gastrointestinal disorders are among the most common outcomes. Ingestion of foreign objects, particularly plastics, is widespread among ruminants, often resulting in rumen impaction, perforation, and sometimes a slow, fatal progression [59, 67]. For smallholder farmers, the loss of even a single animal can represent a substantial economic setback.

Chronic exposure to heavy metals and pesticide residues further exacerbates these risks. Continuous contamination of water and forage leads to bioaccumulation in vital organs, causing liver and kidney damage, neurological impairment, and general debilitation manifested as reduced weight gain and decreased milk production [32, 93]. Reproductive health is also affected, with toxins disrupting hormonal cycles, reducing fertility, causing irregular estrus, and increasing the incidence of abortions [6, 37, 56].

Waste sites also serve as reservoirs for pathogenic bacteria such as *Salmonella*, *E. coli*, and *Leptospira*, which can trigger mastitis, leptospirosis, and other diseases, undermining productivity and requiring costly treatments [51, 66]. External health effects, including dermatitis and hoof rot from effluent contact, result in lameness and restricted mobility [39, 69].

3.4.2 Zoonotic disease transmission mechanisms

The analysis identified multiple, overlapping routes of zoonotic transmission operating at the waste–livestock–human–environment nexus. Direct contact through handling of manure or infected animals remains a major risk for farm and abattoir workers [74]. Indirect pathways include environmental exposure via contaminated soil and water, as well as ingestion of contaminated food products such as milk, meat, and crops irrigated with wastewater. WASH-related fecal–oral routes further amplify transmission in settings with poor hygiene and inadequate sanitation infrastructure [44, 63]. The burden of infection is particularly high among children, pregnant women, and occupationally exposed groups in low- and lower-middle-income countries, where waste segregation and treatment remain limited [66].

Figure 4 illustrates the distribution of zoonotic and AMR risk levels across six major waste-management categories synthesized from the included studies. The synthesis of risk-level data across the reviewed literature revealed distinct gradients of zoonotic disease and antimicrobial resistance (AMR) risks associated with different waste-management factors. Animal manure management consistently demonstrated the highest risk for both zoonotic transmission and AMR emergence. Wastewater disposal, food-chain contamination, and environmental contamination of soil and water were also associated with moderate to high risks, particularly through the dissemination of

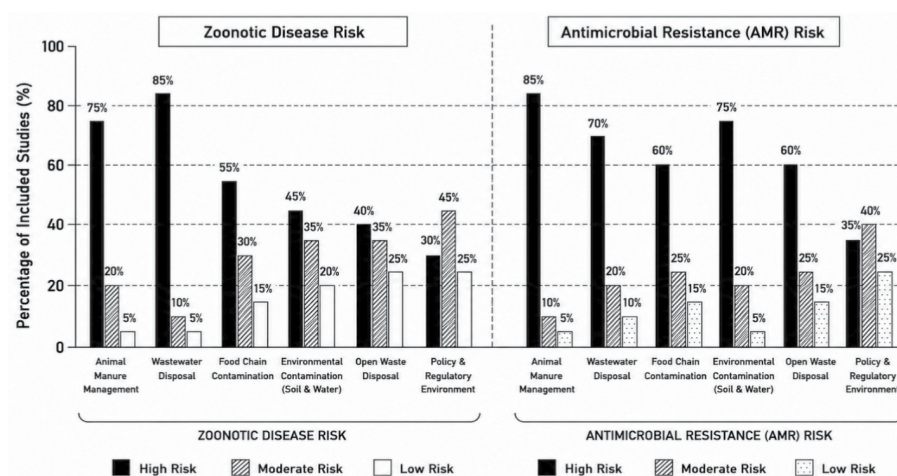


Fig. 4 Distribution of zoonotic and antimicrobial-resistance risk levels across six key waste-management categories derived from included studies

pathogens and antimicrobial-resistance determinants. Open waste disposal contributed to both environmental exposure and indirect disease transmission pathways, while policy and regulatory environments showed variable influence depending on the strength of enforcement and governance systems. Due to methodological differences among included studies, the framework represents a conceptual One Health risk-prioritization approach rather than a quantitative epidemiological risk estimate.

3.4.3 Food safety implications

The consequences of waste contamination extend beyond livestock to human food products, posing a critical public health concern. Table 6 highlights the range of chemical and microbial contaminants identified in meat and milk. Chemical hazards, including heavy metals such as lead and cadmium, originate from improperly disposed batteries, mining residues, and plastics, and can accumulate in animal tissues.

Studies in Ethiopia and India have reported lead concentrations in goat meat and milk exceeding international safety limits [43, 67]. Persistent organic pollutants, including dioxins from burning plastics, accumulate in animal fat and dairy products, with long-term risks such as carcinogenicity and developmental disorders [19, 20, 83]. Microbial contamination presents an immediate threat. Pathogens such as *E. coli* O157:H7 and *Salmonella*, common in waste-rich environments, can contaminate carcasses during slaughter and raw milk during handling. In Zimbabwe, up to 25% of raw milk samples collected near open waste sites tested positive for *E. coli* O157:H7, illustrating a direct pathway from environmental pollution to human consumption [45, 51].

Antimicrobial resistance (AMR) is an emerging concern. Improper disposal of pharmaceuticals and veterinary drugs introduces antibiotic residues into the environment. Continuous exposure of livestock to sub-therapeutic drug levels encourages the development of resistant bacteria, which can be transmitted to humans through meat and milk, reducing the effectiveness of essential medications [68].

The socioeconomic implications are significant. Contamination undermines consumer confidence, restricts access to formal markets, and perpetuates poverty. Rejection of milk or carcasses can severely affect household incomes [89]. However, targeted interventions can mitigate these risks. Training programs in Southern Africa that focus on milk hygiene, proper container cleaning, and safe storage have successfully reduced

Table 6 Common contaminants in meat and milk from waste-exposed livestock

Contaminant type	Likely source	Detected In	Health risk to humans	Citation
Lead (Pb)	Dumped batteries, mining waste	Liver, kidney, milk	Neurotoxicity, cognitive deficits	Mofokeng et al. [48]
Cadmium (Cd)	Plastic waste, fertilizers	Muscle, liver, milk	Renal failure, cancer	Evode et al. [16], Gomaa et al. [24]
<i>E. coli</i> O157:H7	Human and animal feces	Raw milk	Diarrhea, hemolytic uremic syndrome	Chimuti, Mugadza, Ntuli, & Njage (2024) [11]
Antibiotic residues	Improper drug use, no withdrawal period	Milk, meat	Resistance, allergies, and hormonal imbalance	Woolhouse et al. [88]
Mercury (Hg)	Electronic waste, river sediment	Milk	Neurotoxicity, developmental delay	Franco et al. [19]
Arsenic (As)	Contaminated water, industrial runoff	Muscle, milk	Carcinogenesis, cardiovascular disease	Barik et al. [6]

microbial contamination [58], demonstrating that the food safety challenge, though serious, is manageable.

3.5 Environmental degradation and ecosystem costs

The impacts of unmanaged waste extend far beyond the farm gate or the dining table. Contamination from solid and liquid waste triggers cascading environmental damage that threatens fundamental life-support systems, soil, water, and air, upon which both rural communities and their livestock depend. What begins as a localized disposal challenge can thus escalate into a systemic threat to regional ecosystem integrity and climate resilience.

Soil degradation is among the most visible consequences. The open dumping of plastics, metals, and chemical containers compromises soil fertility and structure, obstructing root growth and disrupting microbial activity critical for nutrient cycling [42]. In Zimbabwe and Kenya, soils adjacent to dumpsites show elevated lead and cadmium levels, which are absorbed by forage crops and enter the food chain, threatening long-term agricultural productivity [53].

Water pollution is another critical dimension. Liquid waste and agricultural runoff, often rich in nutrients, pathogens, and chemicals, infiltrate rivers, boreholes, and groundwater aquifers [33]. This contamination jeopardizes drinking water and livestock supplies while promoting eutrophication in lakes and dams, causing algal blooms that deplete dissolved oxygen, trigger fish kills, and destabilize aquatic ecosystems [14, 40]. In Malawi, greywater discharged into streams produced coliform counts exceeding WHO standards, illustrating the direct link between household waste practices and broader aquatic degradation [9, 32].

Air quality is also compromised. Open burning of waste releases dioxins, furans, and particulate matter, impairing local air quality and respiratory health for humans and animals [7]. Additionally, unmanaged organic waste and animal manure emit methane, contributing to climate change [23].

Biodiversity loss represents a further environmental cost. Dumpsites attract scavenging wildlife such as monkeys, dogs, and birds, increasing interspecies contact and the risk of zoonotic disease transmission [66]. In aquatic systems, microplastics consumed by fish and waterbirds introduce synthetic materials into natural food webs, threatening biodiversity and ecosystem stability [14, 40].

Despite these challenges, signs of ecological restoration exist. Community-based composting in South Africa and Kenya has demonstrated that organic waste can be converted into a resource, enhancing soil fertility while reducing landfill loads [42]. Nature-based solutions, such as constructed wetlands in Ethiopia, have successfully filtered nutrients from wastewater, using ecological processes to remediate pollution [73]. These approaches go beyond harm mitigation to promote ecological regeneration, demonstrating how effective waste management can enhance ecosystem resilience.

3.6 Solutions and a one-health way forward

The preceding sections trace a concerning trajectory from evolving waste streams and exposure pathways to severe consequences for animal health, food safety, and environmental integrity. While these risks are complex, the same interconnectedness that sustains the problem provides the foundation for its resolution. Evidence from this review

strongly supports an integrated One Health approach as the most effective and sustainable framework for addressing the waste–livestock–human–environment nexus. Solutions require moving beyond fragmented interventions toward strategies that act simultaneously across multiple leverage points.

Central to this transformation is community empowerment and low-cost innovation. Rural communities are active participants and innovators, not passive victims [85]. Practices such as informal plastic recycling in South Africa [42] and farmer-led composting in Botswana [52] provide viable, context-specific models. Scaling these initiatives requires targeted capacity building and awareness programs that integrate local knowledge. Training for farmers, extension officers, and community health workers on waste segregation, exposure risks, and environmental hygiene can drive behavioral change [13, 79, 92]. It is through managing their own waste streams that communities can transition from being recipients of contamination to custodians of ecological sustainability.

Complementing community action is the deployment of appropriate technologies and decentralized infrastructure. Rural solutions must be affordable, low-maintenance, and productive. Small-scale biogas digesters exemplify this approach, converting livestock manure and organic waste into renewable energy while reducing pathogen loads and methane emissions [30, 31, 77]. Protecting water sources through buffer zones, fencing, and low-cost filtration systems can significantly reduce exposure pathways for humans and animals [60, 62, 69]. These interventions reflect circular economy principles, transforming waste from a liability into a resource [50]. Sustained change also depends on enabling policy and governance frameworks. Effective regulation is necessary to control hazardous waste, curb open dumping and burning, and institutionalize safe disposal practices [82, 89]. Policy must integrate waste management with agricultural productivity, animal health, and public health systems. Collaboration between veterinary and environmental health services can enable early detection of contamination-related diseases [13].

Promoting the safe reuse of treated organic waste as fertilizer further restores soil health and closes nutrient loops [41]. As summarized in Table 7, strategic interventions, from community-led waste collection to circular economy policies, are interconnected elements of a unified One Health strategy. Together, they provide a pathway toward resilient, health-promoting rural ecosystems. A Rural Waste-Mediated One Health Risk Prioritization Framework was developed from the integrated synthesis of findings across the included studies (Fig. 5). Important solid and liquid waste sources, predominant livestock exposure routes, related zoonotic and antibiotic resistance risks, and the socioeconomic and governance elements noted in the review are all included into the suggested framework. These elements are organized to show how dangers associated with waste arise, interact, and are influenced by rural contextual factors. The framework, which is based on the One Health approach, highlights the interdependence of human, animal, and environmental health, with livestock disease management, food safety surveillance, household and farm-level waste management, and policy processes acting as interdependent components within a shared system rather than discrete domains.

The paradigm encourages a multimodal prioritization method that takes into account hazard severity, likelihood of exposure, impacted populations, environmental persistence, zoonotic potential, and contribution to risk rather than assuming uniform priority across recognized hazards, particularly, antimicrobial resistance emergence. Physical

Table 7 Strategies for mitigating waste exposure in rural livestock systems

Strategy	Description	Examples/Applications	Benefits/Outcomes
Improving Waste Management Infrastructure	Developing affordable and locally appropriate waste collection, segregation, and disposal systems in rural communities.	Community-led solid waste collection systems in Molepolole, Botswana (Ntsabane <i>et al.</i> [58]); pilot biogas and sanitation projects in rural Ethiopia (Sime [77]).	Reduced environmental contamination; improved sanitation; lowered livestock exposure to waste.
Promoting Waste Segregation and Recycling	Training and awareness programs to separate hazardous and non-hazardous waste at the source; recycling of plastics and metals.	Plastic and metal recycling initiatives in South Africa (Olatunji <i>et al.</i> [65]; Mazhandu <i>et al.</i> [42]); community-led e-waste management in Ghana (Lebbie <i>et al.</i> [36]).	Lower volume of waste; reduced hazardous waste in livestock environments; resource recovery.
Adoption of Environmentally Friendly Treatment Technologies	Use of alternative treatment methods, such as autoclaving, composting, and biogas digesters, that reduce pathogen load and emissions.	Biogas digesters in rural Kenya and Ethiopia for manure treatment (Hossain <i>et al.</i> [30]); composting of organic waste in Tanzania (Kibonde [34]; Gautam <i>et al.</i> [22]).	Reduced methane emissions, renewable energy generation, and safer waste disposal.
Enhancing Water Source Protection	Fencing and buffer zones around water sources, controlled disposal of waste, and simple water treatment techniques.	Greywater reuse and solar disinfection practices in rural South Africa (Radingoana, 2021); low-cost filtration in Tanzania (Okafor <i>et al.</i> [62]).	Improved water quality; reduced contamination risks for livestock and humans.
Capacity Building and Awareness Raising	Education and training of community members, farmers, extension workers, and health personnel on waste risks and management.	Community waste management education in Chegutu, Zimbabwe (Musasa <i>et al.</i> [52]); One Health training for zoonotic risk reduction in Zambia (Martineau [39]).	Behavioural change; improved local capacity; sustainable waste practices.
Policy Development and Enforcement	Creation and enforcement of regulations controlling waste management practices, including hazardous waste handling.	Circular economy integration into waste policy in Namibia and Ghana and Zimbabwe (Zikali <i>et al.</i> [94]; Erasmus <i>et al.</i> [14]; Volsuuri <i>et al.</i> [86]).	Institutionalization of good practices; improved environmental and animal health outcomes.
Integrating Waste Management with Livestock Production	Linking waste reuse (e.g., treated organic fertilizer) with livestock feeding and health monitoring to enhance system resilience.	Organic fertilizer production from livestock waste in Nigeria (Mamman Anzaku [38]); safe compost use in Ugandan peri-urban dairy farms (Jimenez <i>et al.</i> [32]).	Increased pasture productivity; reduced environmental risks; early detection of contamination effects.

waste hazards may differ in importance based on local exposure conditions, but environmental dissemination of antimicrobial resistance determinants and waste-associated zoonotic pathogen transmission may require higher priority due to their wider public health implications and cross-sectoral impacts. Therefore, risk prioritizing is seen as a dynamic process that is influenced by environmental sensitivity in rural areas, animal production systems, waste management techniques, and local epidemiological data. In this sense, the framework is meant to assist researchers, practitioners, and policymakers in recognizing and ranking the hazards associated with waste-mediated zoonotic and antibiotic resistance. Lastly, there are a few constraints that should be considered when interpreting the framework. Its usefulness may differ among rural production systems and geographic locations because it is based on conceptual analysis and integrative evidence synthesis rather than field validation.

Furthermore, methodological variations among included studies, especially in antimicrobial resistance surveillance and environmental exposure assessment, may impact comparability, while heterogeneity in waste management practices, livestock systems, environmental conditions, and institutional capacity may impact the applicability of particular pathways across contexts.

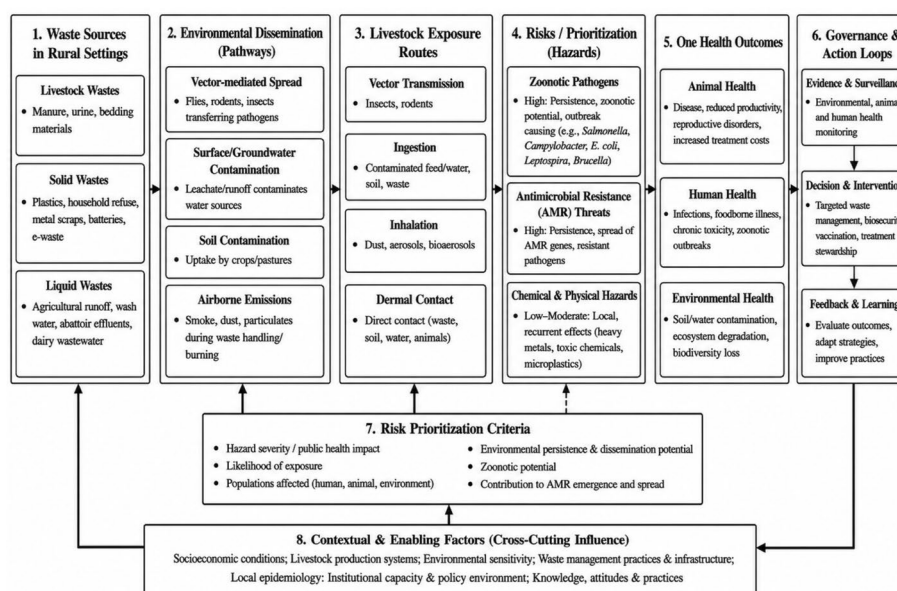


Fig. 5 Rural waste-mediated one health risk prioritization framework developed by the authors based on the thematic synthesis of the reviewed studies

4 Research gaps and future directions

Despite increasing recognition of the waste–livestock–human–environment nexus as a One Health priority, substantial research and policy gaps persist. Most of the available literature remains concentrated in high-income or urban settings, while data from rural low- and middle-income regions, where risks are most acute, are scarce. Empirical evidence from Zimbabwe and other sub-Saharan African contexts is particularly limited, hindering the development of context-appropriate interventions.

First, there is a lack of quantitative microbial risk assessment (QMRA) and longitudinal data capturing seasonal and spatial variations in pathogen and resistance gene prevalence. The majority of studies reviewed were cross-sectional and descriptive, limiting causal inference on how waste exposure translates into human or animal disease outcomes.

Second, AMR surveillance at the environmental interface remains underdeveloped. Few studies simultaneously monitor animal, human, and environmental samples, and those that do often lack standardized laboratory methods or genomic tools for resistance profiling. Integrating molecular surveillance into community-based monitoring could yield more comprehensive insights into resistance pathways.

Third, policy and implementation research gaps hinder translation from science to practice. Most interventions tested, such as composting, wastewater treatment, and biosecurity protocols, lack cost-effectiveness analysis or scalability assessments under rural socioeconomic constraints. There is also minimal evaluation of governance frameworks linking waste management, public health, and veterinary regulation.

Fourth, the social dimensions of risk and behavior remain understudied. Cultural norms around animal–human cohabitation, traditional manure reuse, and perceptions of hygiene profoundly shape exposure patterns but are rarely quantified. Ethnographic and participatory research could illuminate these dynamics and inform community-tailored interventions.

Finally, interdisciplinary methodological integration is still weak. Environmental microbiology, veterinary epidemiology, and rural development research often operate in silos, producing fragmented insights. Future studies should embrace transdisciplinary designs that combine microbial ecology, environmental engineering, social science, and economic modeling to create adaptive, locally grounded solutions.

Future research should therefore prioritize (i) context-specific One Health surveillance frameworks, (ii) harmonized AMR monitoring protocols across animal and environmental compartments, and (iii) participatory risk-reduction strategies that couple technical innovations with community ownership. Such approaches are essential to operationalize the Rural Waste-Mediated One Health Risk Framework proposed in this review and to build resilient rural systems that protect human, animal, and environmental health concurrently.

5 Conclusion

This review shows that waste management, livestock health, and human health are closely interconnected in rural systems. Literature indicates that unmanaged manure, wastewater, and other wastes can contaminate livestock feed and water and contribute to human exposure through food and water pathways, with implications for disease risk, productivity, and environmental quality. The findings suggest that existing rural waste management practices are often insufficient when implemented in isolation, highlighting the need for integrated, context-specific approaches. Guided by a One Health perspective, this study proposes a Rural Waste-Mediated One Health Risk Prioritization Framework to support the identification and ranking of zoonotic and antimicrobial resistance risks at the waste–livestock–human–environmental interface. The framework provides a basis for informing coordinated interventions, surveillance, and policy development in rural settings.

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Author contributions

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Data availability

Data sharing is not applicable to this article as no datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

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